Maternal and child health

Maternal and child health outcomes in low-resource and geographically challenging settings like Nepal remain constrained by delayed care-seeking, limited clinical expertise at the community level, and fragmented referral pathways, often turning preventable conditions into life-threatening emergencies. there are 52000 femele community health workers in Nepal.

We propose an offline-first, artificial intelligence-assisted decision support platform designed specifically for frontline community health workers. The system enables multilingual, voice-based symptom capture, predictive risk stratification, and smart referral recommendations.

By empowering Female Community Health Volunteers with human-in-the-loop intelligence, the platform enables earlier *detection* of high-risk cases, reduces referral delays, and improves maternal and child health outcomes in underserved communities.

Explainable risk scores and referral recommendations are provided to support a human-in-the-loop decision model aligned with existing workflows.

While artificial intelligence has shown potential for maternal and neonatal risk assessment, most existing solutions are facility-centered, connectivity-dependent, or limited to pilot deployments.

An offline-first system architecture ensures functionality in low-connectivity environments, with secure data synchronization when connectivity is available.

The proposed system enables predictive, community-level triage, improves early identification of high-risk pregnancies and childhood illnesses, and reduces referral delays by guiding timely and appropriate care escalation.

This study proposes an AI-assisted decision support framework to augment FCHV-led maternal and child health services.

Ram is a good boy	Ram lives in kathmandu
Ram loves music	Ram is a 13 years old
Yesterday ram was sick	Ram had 103 degree fever

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